INCH-POUND
MIL-PRF-13587F(AT)
14 March 1996
SUPERSEDING
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16 September 1986

PERFORMANCE SPECIFICATION

BIN, STORAGE, AGGREGATE, PORTABLE, 60 TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a one-compartment, 60-ton-capacity, portable aggregate storage bin for use in the loading of trucks.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36	- Structural Steel.
A120	- Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A575	- Steel bars, Carbon Merchant Quality, M-Grades.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Steel Construction Manual.

(Application for copies should be addressed to the American Institute of Steel Construction, 101 Park Avenue, New York NY 10017.)

AMERICAN WELDING SOCIETY INC. (AWS)

D1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC Z1.4-1993 - Sampling, Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2).

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3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for bins. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.4.1). Recovered materials shall be used to the maximum extent practicable.

3.2.1 Material deterioration prevention and control. The bin shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that maybe encountered in any of the applicable operation and storage environment to which the bin maybe exposed. Asbestos, cadmium, and radioactive material will not be used in this item. Radioactive material is defined by Title 10, Code of Federal Regulations, Part 40, and radioactive. material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.2.3 Identification of material and finishes. When specified, the contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall provide this information to the contracting officer or designated representative.

3.2.4 Structural steel. When used, structural steel shall conform to ASTM A36.

3.2.5 Steelbar and pipe. When used, steel bar shall conform to ASTM A575 and pipe shall conform to ASTM Al 20.

3.3 Desire and construction. The bin shall be of the bolted, knockdown single-compartment type with a gate at the bottom for loading trucks (see 4.4.2). The bin shall be provided with four supporting columns and with support braces located in such a manner to create a passageway under the bin with the sides formed by cross bracing. Lateral clearance of the passageway shall be not less than 11 feet, and vertical clearance shall be not less than 10 feet from the lowest point of the bin gate to the top of the base plates. The inside dimensions of the bin measured at the top, shall be not more than 12 feet, 10 inches square. The slope of the sides of the bin hopper shall be not less than 45 degrees from the horizontal. With the gate fully opened, the bin shall discharge completely when loaded with aggregate.

3.3.1 Capacity. The bin shall have a heaped capacity of not less than 60 tons based on aggregate weighing not less than 100 pounds per cubic foot by commercial standards. When loaded to capacity, the bin shall show no visual evidence of distortion or permanent deformation of any part, member, or supporting structure or failure of any welded or bolted connection (see 4.4.3 2).

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3.3.2 Bin body. The bin sidewalls shall be made of steel not less than 3/16 inch in nominal thickness and the sloping bottom plates shall be not less than 1/4 inch in nominal thickness or equivalent (see 4.4.2).

3.3.3 Bin columns. Columns shall be constructed in accordance with the requirements of the AISC Steel Construction Manual to withstand the weight of the bin loaded to maximum capacity and the weight of the structural members when subjected to a wind load of not less than 20 pounds per square foot. Each column shall be provided with a base plate having not less than two holes to accommodate bolts not less than 1 inch in diameter (see 4.4.2).

3.3.4 Support braces and bin stiffeners. Support braces and bin stiffeners shall be provided and located to provide a safety factor of not less than 1.5 based on the minimum yield stress (for materials that do not exhibit a well defined yield stress, 0.002 strain offset shall be used to establish the yield stress) of the material used (see 4.4.2).

3.3.5 Bolts. Bolts used in the construction of the bin and for fastening the braces shall be provided and spaced so as to have a safety factor of 4 in shear based on the minimum ultimate tensile strength of the material used for the bolts and with the bin filled to maximum capacity (see 4.4.2).

3.3.6 Bin gate. The bin gate shall be of the single or double-clam type constructed of cast steel, fabricated from structural steel or equivalent. The clear opening of the bin gate shall be not less than 200 square inches (see 4.4.2).

3.3.7 Bin gate manual control. A manual control shall be provided for the bin gate which shall permit a variable gate opening between wide open and closed. The control shall be accessible to an operator standing on the ground at a safe distance from bin discharge and shall provide sufficient leverage to open and close the gate on the fully loaded bin. The control shall be labeled “GATE CONTROL” and shall be provided with a double-ended arrow showing direction of operation and labeled at each end (OPEN and CLOSE) to indicate the functional result (see 4.4.2 and 4.4.3.3).

3.3.8 Erection drawings and assembly/disassemble. Two complete sets of step-by-step erection drawings shall be furnished with each bin. Each part shall be identified with a part number corresponding to the part number on the erection drawings. The bin shall be assembled and disassembled without being cut or welded and each section of the bin weighing 100 or more pounds shall be provided with removable lifting attachments. Disassembly shall be accomplished in not more than 6 hours, with a total of not more than 24 manhours When disassembled, no part of the bin shall exceed 13 feet in length and 8 feet in width (see 4.4.3.4)

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3.4 Identification and marking. Identification and marking shall be permanent and legible. Marking shall be applied to a nameplate attached to the bin by screws, bolts, or rivets in a conspicuous protected location (see 4.4.2). The nameplate shall be made of a material suitable to withstand environmental conditions expected with a aggregate storage bin. The bin nameplate shall inch-de the following information:

a. Bin, storage, aggregate, portable, 60 Ton
b. National stock number (NSN)
c. Military part number
d. Manufacturer's serial number
e. Manufacturer's name or identification code (CAGE)
f. Contractor or order number
g. Date of manufacture
h. US

3.5 Treatment and painting. The portions of the storage bin normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practices (see 4.4.2). The final top paint coat color shall be as specified in the contract or order (see 6.2). When specified (see 6.2), the storage bin shall then be overcoated with camouflage pattern. If chemical agent resistant coating (CARC) is required, it shall be specified in the contract or order (see 6.2).

3.6 Safety. The storage bin shall be provided with a grating as means of guarding the top opening. The grating shall not materially obstruct the opening but will be substantially constructed to prevent persons from falling in (see 4.4.2).

3.7 Workmanship. All parts, components, and assemblies of the storage bins including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs and sharp edges (see 4.4.2).

3.7.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter (see 4.4.2). Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be in accordance with AWS D1.1 Structural Welding Code - Steel. Welds shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in the weld metal, lack of fusion with parent metal or between parts, incomplete penetration, lack of prescribed fit, or spatter (see 4.4.2).

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4. VERIFICATION

4.1 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.21 Examination. The first article bin shall be examined for defects as specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.2.2 Test. The first article bin shall be tested as specified in 4.4.3.1 through 4.4.3.4. Failure of any test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Examination. Each bin shall be examined for defects as specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.4 Method of inspection.

4.4.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Defects. Conformance to 3.2, 3.2.1, 3.2.2, 3.3.3 thru 3.3.8 and 3.4 thru 3.7.1 shall be determined by examination for the defects listed in table I. Examination shall be visual, tactile, or by measurement with SIE.

4.4.3 Tests. The bin shall be tested as follows.

4.4.3.1 Assembly. Erect the bin in accordance with the erection drawings (see 3.3.8). A suitable crane shall be used during assembly. Eyes provided on the bin components shall be used during assembly. Measure the lateral and Vertical clearance of the passageway. Inability to erect the bin in accordance with the step-by-step procedures on the drawings (see 3.3.8), clearance not as specified in 3.3 or any permanent damage to the lifting eves shall constitute failure of this test

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TABLE 1. Classification of defects.

Category	Defect	Method of examination
Major:		
101	Dimensions not as specified (see 3.3).	SIE 1/
102	Material not as specified (see 3.2).	Visual
103	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.2.1).	Visual
104	Dissimilar metals are not effectively insulated from each other (see 3.2.2).	Visual
105	Columns not constructed as specified (see 3.3.3).	Visual
106	Base plate not as specified (see 3.3.3).	Visual
107	Safety not as specified (see 3.6).	Visual
108	Gate not single- or double-clam type (see 3.3.6).	Visual
109	Bin gate opening less than 200 square inches (see 3.3.6).	SIE 1/
110	Manual control to bin gate not as specified or not accessible (see 3.3.7).	Visual
111	Two complete sets of erection drawings not furnished (see 3.3.8).	Visual
112	Assembly instructions not complete or not correct (see 3.3.8).	Visual
113	Parts or components missing or do not function (see 3.3.8).	Visual
114	Identification marking not as specified (see 3.4).	Visual
115	Lifting attachments missing (see 3.3.8).	Visual
116	Treatment and painting not as specified (see 3.5).	Visual
117	Color not as specified (see 3.5).	Visual
118	Camouflage patterns not as specified (see 3.5).	Visual
119	Workmanship not as specified (see 3.7).	Visual
120	Welding not as specified (see 3.7.1).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3.2 Loading. Load the bin with 60 tons of aggregate weighing the equivalent of not less than 100 pounds per cubic feet. Inability of the bin to hold the specified capacity, or visual evidence of distortion or permanent deformation of any part, member, supporting structure, or welded or bolted connection shall constitute failure of this test (see 3.3.1).

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4.4.3.3 Operational. Drive trucks under the loading-bin gate. Open and close the gate as each truck is loaded until the bin is completely, empty of aggregate. Inability of the bin gate to be fill-opened or closed by one operator, or failure of the bin to discharge (without aid) all the aggregate when the gate is open shall constitute failure of this test (see 3.3.7).

4.4.3.4 Sectionalization. Disassemble the bin after completion of the operational test using a suitable crane and the lifting eyes provided on the applicable bin components. Failure of the bin to be disassembled within 6 hours, with a total of not more than 24 manhours, into parts not more than 13 feet long nor more than 8 feet wide; or any permanent damage to the lifting eyes or to the components to which they are attached shall constitute failure of this test (see 3.3.8).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a genera! or explanatory nature which maybe helpful, but is not mandatory.)

6.1 Intended use. The aggregate storage bin is intended for use during the construction of military roads and airfields.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. When first article is required (see 3.1).
d. Color of paint final top coat (see 3.5).
e. If storage bin is to be overcoated with camouflage paint pattern (see 3.5).
f. If CARC paint is required (see 3.5).
g. Packaging requirements (see 5.1).

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6.3 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of bins.

6.4 Subject term (key word) listing.

Body
Braces
columns
Gate
Hopper
Loading

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Review Activity:
Army - MT

Preparing Activity:
Army - AT

(Project 3895-0001)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-13587F(AT)	960314

3. DOCUMENT TITLE
Bin, Storage, Aggregate, Portable, 60 Ton

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(810) 574-8745	786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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